Table 1: Data privacy (accuracy↓) comparison when using robust regression under different perturbation budgets β $(0-8)$, where $\beta=0$ indicates no defense measures.

Method	Perturbation Budget β								
	0	1	2	3	4	5	6	7	8
PoiGen	0.7644	**0.5635**	0.5529	0.5591	0.5512	0.5529	0.5499	0.5506	0.5520
BREW	0.7644	0.6893	0.6504	0.6417	0.6404	0.6605	0.6362	0.6251	0.6242
MaxEnt	0.7644	0.7676	0.7693	0.7705	0.7718	0.7726	0.7715	0.7718	0.7708
ARL	0.7644	0.7678	0.7667	0.7654	0.7621	0.7723	0.7676	0.7696	0.7705
Ours	0.7644	0.5983	**0.5222**	**0.5230**	**0.5215**	**0.5150**	**0.5204**	**0.5163**	**0.5230**

Table 2: Performance comparison when modifying existing behaviors. For privacy evaluation, accuracy ↓ is used. For utility evaluation, RMSE ↑ is used in Google Play and accuracy ↑ is used in Weibo.

Method	Data Privacy								Data Utility			
	Google Play				Weibo				Google Play		Weibo	
	LR	DNN	SVD+LR	RF	LR	DNN	SVD+LR	RF	MF	Slope One	GCN	GAT
Attack	0.4426	0.4653	0.3300	0.2734	0.8073	0.8120	0.7843	0.7224	0.2400	0.2871	0.6868	0.7037
PoiGen	0.3990	0.4200	0.2829	0.2078	0.7940	0.6190	**0.5655**	0.5845	0.2229	0.2575	0.6881	0.7047
BREW	0.2669	0.2759	0.2214	0.1321	0.7376	0.7275	0.7332	0.6922	0.2218	0.2588	0.6835	0.7053
MaxEnt	0.4083	0.4061	0.2534	0.0887	0.7676	0.8039	0.5568	0.5240	0.2088	0.2270	**0.6900**	0.7036
ARL	0.4243	0.4474	0.2672	0.1345	0.8040	0.8106	0.7316	0.7047	0.2096	0.2217	0.6894	**0.7058**
Ours	**0.0597**	**0.0822**	**0.0706**	**0.0760**	**0.5704**	**0.4914**	0.5675	**0.5436**	**0.2353**	**0.2915**	0.6887	0.7028

Table 3: Data privacy comparison when existing multi-categorical attributes on Google Play. This is done by discretizing the rating of each product into unrated, low-rated (<= 0.6), and high-rated (>= 0.8) so that each rating feature is a three-dimensional one-hot vector.

Method	Accuracy ↓				Loss ↑		
	LR	DNN	SVD+LR	RF	LR	DNN	SVD+LR
Attack	0.3458	0.2993	0.1296	0.2663	3.4850	3.4213	3.9472
PoiGen	0.3174	0.2796	0.1278	0.1884	3.4656	3.3639	3.9410
BREW	0.2943	0.1715	0.1084	0.0680	3.4907	3.8950	3.9611
MaxEnt	0.3436	0.3297	0.1364	0.2762	3.4876	3.2566	3.9307
ARL	0.3427	0.3100	0.1308	0.2746	3.4868	3.3498	3.9345
Ours	**0.1829**	**0.1262**	**0.1075**	**0.0954**	**3.5907**	**4.7783**	**3.9797**

Table 4: Data privacy comparison on the Extended Yale B dataset. Here we restrict the L_∞ norm of perturbation and set β to 0.05

Method	Accuracy ↓				Loss ↑		
	LR	DNN	SVD+LR	RF	LR	DNN	SVD+LR
Attack	0.5359	0.8111	0.4066	0.3614	2.7409	1.1387	2.8926
PoiGen	0.5133	0.7556	0.3963	0.0965	2.9030	**1.5557**	3.0424
BREW	0.4846	0.7885	0.3634	0.2444	2.7552	1.2241	3.0550
MaxEnt	0.6366	0.8583	0.5606	0.4887	2.4438	0.9889	2.4481
ARL	0.5483	0.8296	0.3963	0.3552	2.6756	1.1020	2.8110
Ours	**0.4045**	**0.7310**	**0.2957**	**0.0575**	**2.9154**	1.4960	**3.3440**

Table 5: Data utility (Accuracy ↑) comparison on Extended Yale B dataset. Here we restrict the L_∞ norm of perturbation and set β to 0.05. The target task model uses a three-layer neural network.

No defense	PoiGen	BREW	MaxEnt	ARL	Ours
0.9302	0.8789	0.9299	0.9055	0.8932	0.924